Structural Disruption of Mucin by PET Nanoplastics and Its Consequences on *Escherichia coli* Physiology

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Introduction

Micro- and nanoplastics (MNPs) are pervasive environmental contaminants increasingly recognized for their capacity to interact with biological interfaces and alter microbial behavior. Polyethylene terephthalate (PET), one of the most common polymers used in food and beverage packaging, is therefore relevant to the human gastrointestinal tract, where plastic particles may encounter mucus before reaching epithelial surfaces.

Mucin glycoproteins form the hydrated gel layer that constitutes the primary gastrointestinal barrier. Because this network regulates bacterial adhesion, colonization, and biofilm formation, physicochemical changes induced by MNPs may weaken barrier function and reshape mucosal microbial responses. This study examined PET MNP–mucin interactions using ultraviolet–visible (UV–Vis) absorption, fluorescence spectroscopy, and molecular docking, then evaluated how PET-modified mucin influenced *Escherichia coli* physiology.

By altering mucin before bacteria encounter it, PET nanoplastics may disturb mucosal ecosystems through an indirect barrier-mediated mechanism rather than through direct bacterial exposure alone.

PET MNPs → **Mucin network** → ***E. coli* response**

Materials & Methods

Porcine stomach mucin was prepared at 0.1, 1.0, and 5.0 mg/mL and exposed to in-house spherical PET nanoplastics at 2, 10, and 20 µg/mL, denoted 1×, 5×, and 10×, respectively. Mixtures were incubated in phosphate buffer at 37 °C for 1 h to simulate short mucosal contact.

Design factor	Level 1	Level 2	Level 3
Mucin	0.1 mg/mL	1.0 mg/mL	5.0 mg/mL
PET dose	2 µg/mL (1×)	10 µg/mL (5×)	20 µg/mL (10×)

All exposures: phosphate buffer, 37 °C, 1 h.

Optical changes were assessed by UV–Vis spectroscopy, including absorbance shifts and indices associated with humification and aromaticity. Fluorescence spectroscopy was used to evaluate quenching behavior and local rearrangement within the mucin environment after PET contact.

A three-dimensional porcine mucin-2 model was generated with SWISS-MODEL using human MUC2 as the template. Molecular docking of a PET trimer (3PET) was then performed with AutoDock-GPU to map binding affinity and interaction hotspots. For microbial assays, *E. coli* ATCC 35218 was incubated for 24 h at 37 °C with untreated or PET-treated mucin. Bacterial biomass was estimated by optical density at 600 nm, and biofilm formation was quantified by crystal-violet staining at 595 nm.

PET–mucin contact → **UV–Vis + fluorescence** → **3PET docking** → ***E. coli* assays**

Study workflow: exposure → characterization → docking → microbial assays.

Results — Structural Alterations

PET exposure produced concentration-dependent remodeling of the mucin network. At low-to-intermediate mucin concentrations, progressive increases in aromaticity and humification indices were accompanied by mild fluorescence enhancement, suggesting local microenvironmental changes and re-exposure of aromatic residues that had previously been masked within the glycoprotein matrix.

Dense mucin behaved differently. At 5.0 mg/mL, the network showed a buffering response characterized by low humification, slight fluorescence quenching, and reduced absorbance after PET exposure. This pattern is consistent with PET-induced compaction rather than uniform loosening across all mucin concentrations.

Mucin state	Spectroscopic interpretation
0.1–1.0 mg/mL	Increased aromaticity and humification with mild fluorescence enhancement.
5.0 mg/mL	Buffering behavior with compaction-associated quenching and reduced absorbance.

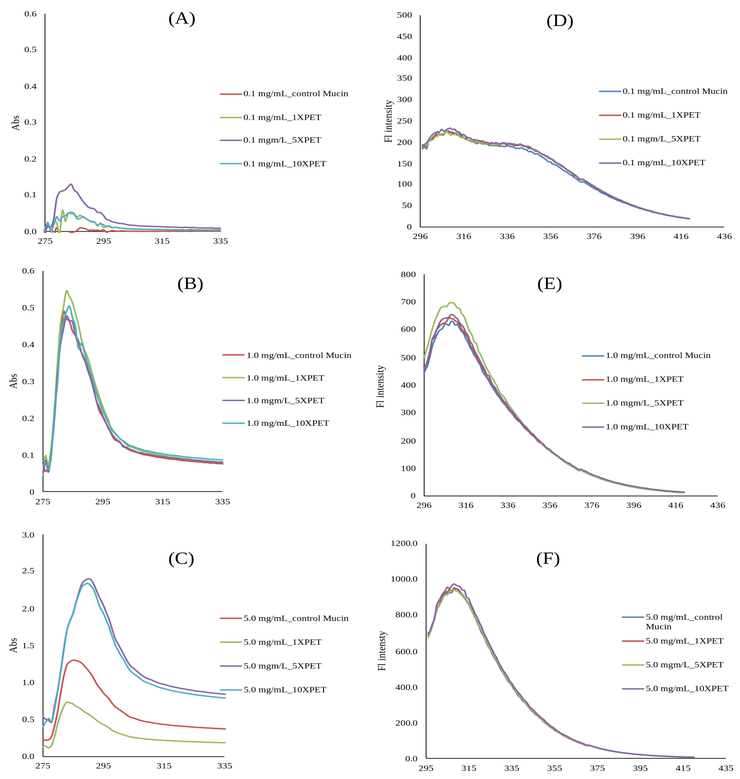


Figure 1. UV–Vis spectra (A–C) and fluorescence spectra (D–F) of mucin at 0.1, 1.0, and 5.0 mg/mL before and after exposure to PET MNPs at 1×, 5×, and 10×.

Results — *E. coli* Physiology

The structural state of mucin governed the downstream bacterial response. Intermediate mucin conditions favored increased *E. coli* biomass accumulation after PET treatment, whereas dense mucin networks limited growth, indicating that the biological outcome depended on the physical state of the barrier rather than PET dose alone.

At 5.0 mg/mL mucin, PET exposure suppressed biofilm formation below untreated-mucin controls. The tightened network is consistent with a physical constraint on the surface attachment required for colonization. Oxidative-stress readouts, including reactive oxygen species (ROS), lipid peroxidation (LPO), glutathione reductase (GR), and superoxide dismutase (SOD), further indicated concentration-dependent modulation of bacterial stress responses.

Barrier context	Physiological outcome
Intermediate mucin	Increased biomass accumulation after PET treatment.
Dense mucin	Reduced growth and suppressed biofilm formation relative to untreated mucin controls.

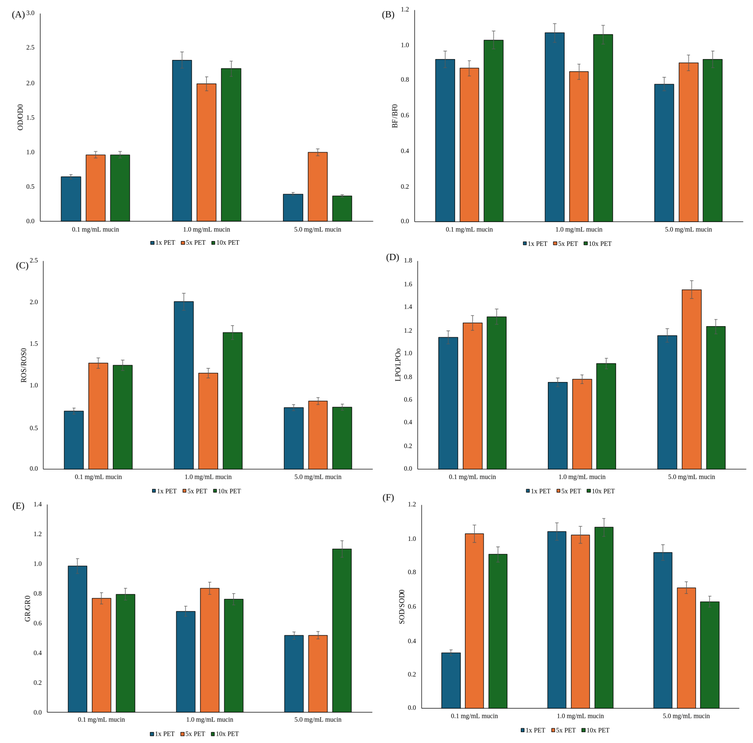


Figure 3. Effect of PET-treated mucin on E. coli physiology: ratios for optical density (A), biofilm formation (B), ROS (C), LPO (D), GR activity (E), and SOD activity (F), normalized to untreated mucin controls.

Results — Molecular Docking

Molecular docking supported a stable association between 3PET and porcine mucin-2, with a predicted binding affinity of −10.33 kcal/mol. The binding pose indicated preferential interaction with a cysteine-rich functional region, consistent with a localized site where PET could interfere with higher-order mucin assembly.

Occupation of this domain may create a steric barrier to the intermolecular disulfide bonding required for mucin polymerization. This mechanism provides a plausible molecular basis for the experimentally observed changes in mucin network organization and barrier integrity.

Predicted interaction: 3PET association with a cysteine-rich mucin region that is important for polymer network formation.

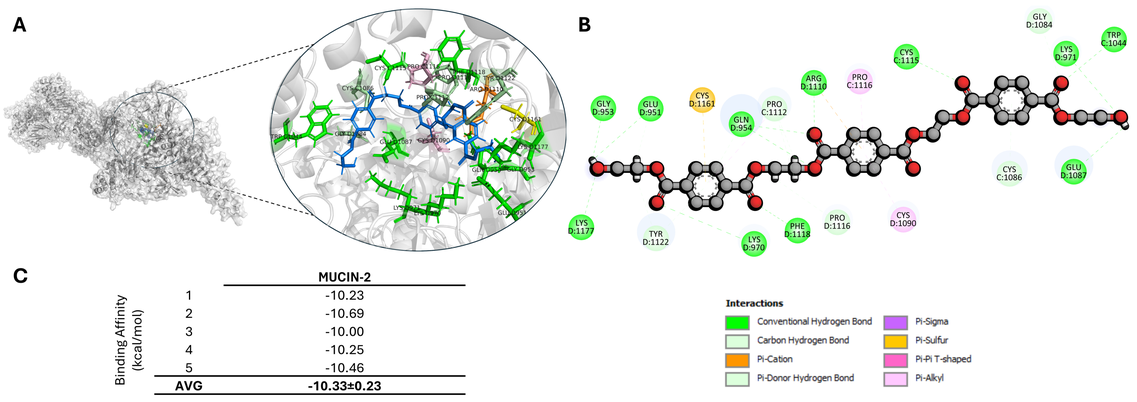


Figure 2. Molecular docking between porcine MUCIN-2 and the PET trimer (3PET): 3D binding pose (A), 2D non-covalent interaction diagram (B), and summary of predicted binding affinities (C).

Conclusions

1. Experimental and computational analyses together indicate that PET micro/nanoplastics **bind stably to mucin and alter its structure** in a concentration-dependent manner.
2. Docking supports **preferential PET binding at structurally important cysteine-rich domains**, providing a plausible molecular basis for impaired mucin polymerization.
3. PET-modified mucin **indirectly influences *E. coli* physiology**, modulating growth and biofilm formation.
4. The findings highlight **a mechanism by which environmental plastics may compromise mucosal ecosystems** by weakening their primary biological barrier.

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